Latent Bridge Matching: IDT-Calibrated Latent Transport for Domain-Level Artistic Transfer

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Abstract

Domain-level artistic transfer has a blunt evaluation failure: when the source is already art, doing nothing can score as target style. On a CLIP-separated WikiArt split, the reproduced SaMAM run remains below the identical-image transfer-only CLIP-S floor through its last trusted 2250-step checkpoint. We introduce *Latent Bridge Matching* (LBM), a 3.9M-parameter latent-transport model that selects target-domain endpoints during training, executes a style-conditioned vector field, and applies terminal distribution matching in VAE latent space. On Distinct5-512, LBM-F/K exceed IDT by +0.0244/+0.0312 transfer-only CLIP-S after 1.2 minutes of checkpoint-training wall time; SaMST clears the same floor only with much larger displacement and high targetwise ArtFID. On the historical strict-750 reference surface, LBM reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image, motivating a simple rule for style-transfer reporting: measure the unchanged-image floor before treating raw target-style affinity as stylization.

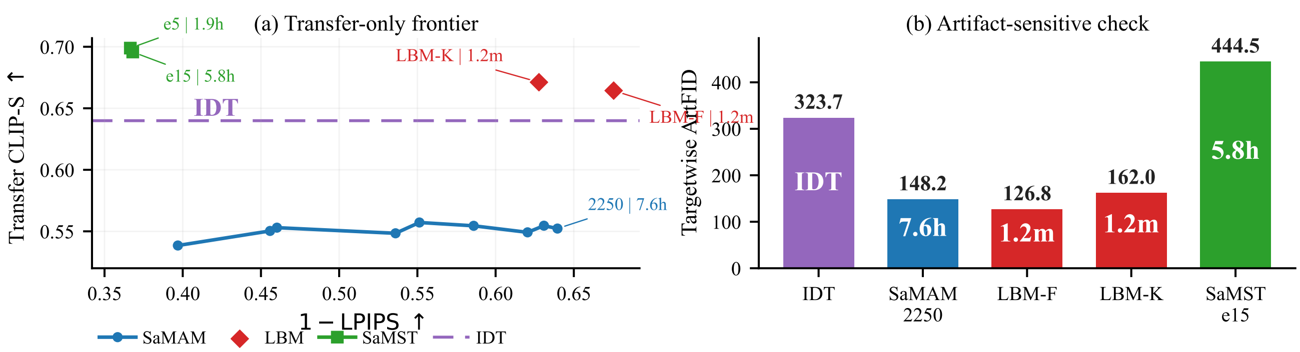

Figure 1: IDT-calibrated Distinct5-512 transfer-only results: SaMAM edits without beating IDT, SaMST beats IDT with large displacement and high targetwise ArtFID, and LBM-F/K beat IDT in a lower-damage region.

Introduction

Domain-level artistic transfer asks for a stylized image from a source image and a target style identifier. This is a compact deployment setting: the model need not receive a reference style image or run per-image optimization at inference. It also exposes a failure in common evaluation practice. When both source and target domains are artworks, an unchanged source can already obtain a strong target-style score. A method can therefore appear competitive by preserving an art-domain prior, or by making visible but misdirected edits, without moving toward the requested target domain. The first question should be signed target-style movement over the unchanged-image baseline, not raw target-style affinity.

This failure is not the familiar style/content trade-off. Classical neural style transfer introduced feature reconstruction and Gram statistics (Gatys, Ecker, and Bethge 2016); feed-forward and attention-based arbitrary style transfer improved efficiency and local correspondence (Huang and Belongie 2017; Park and Lee 2019; Deng et al. 2022; Zheng et al. 2024). Recent systems now span reference-guided stylization (Hong et al. 2023; Huang et al. 2024; Zhang et al. 2025; Shang et al. 2025), compact multi-style models selected by style identity (Liu et al. 2024), and training-free or diffusion-prior methods (Gim, Yoo, and Uh 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024; Jiang et al. 2025; Xu et al. 2025). Across these regimes, automatic metrics can conflate three different events: preserving a source that is already artistic, producing plausible art-domain texture, and moving toward the requested target style. In the style-id setting, the distinction is decisive because inference receives only a domain label.

The representation problem is geometric as well as statistical. VAE latents are compressed feature coordinates with nontrivial local curvature, not RGB pixels. Treating target style as pointwise Euclidean endpoint reconstruction can encourage mean solutions, suppress semantically valid local rearrangements, and blur the line between preserving content and doing nothing. We instead model stylization as controlled latent transport: target-domain examples select where supervision should pull, a style-conditioned vector field decides how to move, and terminal distribution matching checks whether the integrated endpoint has target-style patch statistics.

We propose *Latent Bridge Matching* (LBM), a compact latent-transport model built around that evaluation requirement. LBM combines (i) training-side target-domain endpoint selection for unpaired supervision; (ii) a time-conditioned latent vector field for executable transport; and (iii) SA-SWD, a terminal regularizer that pulls the Euler-integrated endpoint toward target-style patch statistics using projection axes selected from semantic routing. A learnable style spatial prior and semantic cross-attention provide domain-level conditioning without per-image optimization or diffusion sampling.

Our contributions are:

- **We introduce LBM, a compact latent-transport design for domain-level artistic transfer**, combining target-domain endpoint selection, style-conditioned vector-field execution, and terminal distribution matching as an alternative to direct latent endpoint reconstruction.
- **We expose the identical-image prior as a required control**, showing on Distinct5-512 that a reproduced baseline can visibly edit images yet score below IDT in transfer-only CLIP-S.
- **We report compact operating-point evidence with explicit cost**: LBM clears the Distinct5-512 IDT floor after 1.2 minutes of checkpoint-training wall time in the reproduced setup, and reaches 0.7161 CLIP-S / 0.4514 LPIPS at 114 ms/image on the historical strict-750 surface. Matched controls tie weaker kinetic regularization to larger executed motion and a worse style/content trade-off.

Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses (Gatys, Ecker, and Bethge 2016). AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics (Huang and Belongie 2017). Attention-based methods such as SANet and StyTr2 improved local style matching and long-range interaction through learned content-style correspondence (Park and Lee 2019; Deng et al. 2022). More recent reference-guided arbitrary stylization continues to refine the injection path itself: Puff-Net emphasizes cleaner content/style feature separation, HSI replaces point-wise attention with a holistic injector, and SCSA focuses on semantic-region consistency in arbitrary semantic style transfer (Zheng et al. 2024; Zhang et al. 2025; Shang et al. 2025). These works mainly ask how a style exemplar should be fused at inference time. Our problem is different: the main protocol in this paper does not provide a target style image at inference, so the central question is how a domain-level style control should be executed without exemplar-conditioned correspondence.

Efficient multi-style transfer and compact style representations. Multi-style transfer amortizes many styles into one compact model rather than solving a fresh arbitrary-style problem for every target image. SaMST is the closest recent representative: it separates style modeling from style transfer through pluggable style representations and a style-aware transfer network, explicitly targeting the quality-efficiency trade-off for many-style deployment (Liu et al. 2024). This line is the most relevant tokenizer-side baseline for our work. Throughout this paper, "style tokenizer" means a style-id-conditioned control map rather than an image tokenizer, a target-image encoder, or a diffusion-token controller. Our contribution is not another pluggable codebook alone. Instead, we ask whether a compact style representation remains executable after it passes through a latent transport field, so that style-side distinctions survive downstream rendering strongly enough to produce real style movement beyond the unchanged-image baseline.

State-space and Mamba-based style transfer. Recent work also revisits style transfer through state-space backbones. Mamba-ST and SaMam argue that linear-complexity state-space operators can recover global receptive fields more efficiently than quadratic attention, improving the efficiency-quality frontier of arbitrary or many-style transfer (Botti et al. 2025; Liu et al. 2025). These papers are important because they shift the architectural question from "CNN or transformer" to "which global-mixing primitive is most efficient." Our comparisons to SaMam take this line seriously. The distinction is that LBM does not claim a new global-mixing backbone; its main contribution is objective-side: OT-coupled endpoint construction, transport-field learning, and semantic-aligned terminal distribution matching.

Training-free and diffusion methods. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free stylization (Gim, Yoo, and Uh 2024). StyleID and Z* show that large diffusion priors can be adapted for artistic style transfer without retraining by manipulating attention features or attention weights at inference time (Chung, Hyun, and Heo 2024; Deng et al. 2024). SMS further pushes this direction by casting stylization as diffusion-side style distribution matching with additional spectrum-aware and semantic-aware refinement (Jiang et al. 2025). StyleSSP sharpens the same training-free diffusion line by improving the sampling start point to reduce layout drift and style-image content leakage (Xu et al. 2025). These methods can be visually strong, but they generally assume a style reference image and often inherit the inference cost or calibration issues of large generative priors. Our work instead targets compact retrainable latent integration with style-id conditioning rather than iterative diffusion sampling, prompt engineering, or per-style inference-time optimization.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems (Cuturi 2013; Peyré and Cuturi 2019), while the Schrodinger problem connects entropic transport to stochastic path measures (Léonard 2014). We use SWD as a lightweight terminal distribution matching objective and specialize it with semantic routing in SA-SWD. OT serves as a mini-batch endpoint-construction mechanism in latent space, while the formal analysis below characterizes the bounded behavior of the resulting transport objective rather than asserting exact global transport optimality. Evaluation itself remains unsettled in arbitrary style transfer: prior studies report inconsistent protocols across papers and show that commonly used automatic metrics are style-dependent and require calibration to human preference (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our paper builds on that critique but makes a narrower claim. We do not propose a universal replacement metric. Instead, for separated art-to-art transfer we make the unchanged-image prior explicit through an 'idt' reference, then interpret CLIP-derived scores as incremental style movement rather than standalone proof of stylization. This 'idt' baseline is introduced here as a paper-specific reporting diagnostic layered on top of prior metric-caution literature, not as a claim that the exact protocol is already community-standard practice. We therefore combine perceptual and distributional measures including LPIPS (Zhang et al. 2018), FID (Heusel et al. 2017), KID (Bińkowski et al. 2018), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), MANIQA (Yang et al. 2022), and ArtFID (Wright and Ommer 2022). This broader metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts or may partially reflect art-domain priors already present before stylization.

Method

Overview

LBM is best understood as a structured latent transport design for domain-level artistic transfer. Let $z_0 \in \mathbb{R}^{4\times32\times32}$ denote the VAE latent of a content image and let s denote a target style domain identifier. Across the codebase and reproduced checkpoints, the shared design has four coupled components: (i) an optimal-transport coupling that anchors unpaired content and target-style endpoints within each minibatch; (ii) a time-conditioned transport field $v_\theta(z_t, t, s)$ that predicts latent updates; (iii) kinetic regularization that discourages unnecessary latent displacement; and (iv) SA-SWD, which acts as a terminal corrective force pulling the integrated endpoint toward target patch statistics along semantically meaningful axes. At inference, the learned vector field is integrated with a fixed-step solver and decoded by the VAE decoder. In this decomposition, the tokenizer is a semantic beacon that specifies the target region of style space, and the backbone is the executor that realizes the corresponding transport field.

The paper reports two closely related training lineages. The historical bridge formulation uses random-time stochastic-bridge supervision for velocity prediction. The current Distinct5 tokenizer family, however, resolves through the codebase config loader to the OMF path (`objective_mode=omf`), whose active loss is better read as OMF-style endpoint-delta training with terminal matching and kinetic regularization. To avoid overstating closure, we make mechanism claims at the shared transport-design level unless a specific objective family is named explicitly.

Unpaired latent-domain sampling and OT coupling

We train directly on precomputed VAE latents without paired supervision. Each minibatch samples content latents $\{z_c\}$ from all source domains and target-style latents $\{z_s\}$ from the requested target domain. Identity pairs are naturally included through uniform target-domain sampling, which stabilizes the learned transport around same-domain mappings.

Rather than using random target assignments, we construct a mini-batch transport coupling using the sliced-Wasserstein cost. For a batch of content latents $\{z_c^{(i)}\}$ and target-style latents $\{z_s^{(j)}\}$, the SWD cost matrix is

$$C_{ij} = \mathrm{SWD}\big(z_c^{(i)}, z_s^{(j)}\big). \tag{1}$$

An entropic optimal-transport plan is then obtained via the Sinkhorn algorithm:

$$\pi^\star = \arg\min_\pi \sum_{ij} C_{ij}\pi_{ij} + \epsilon H(\pi), \tag{2}$$

where $H(\pi)$ is the entropy regularizer. Each content latent is matched to its coupled target $\tilde{z}_1 \sim \pi^\star(z_s|z_c)$. This coupling provides style-domain endpoints without requiring spatially aligned targets.

Time-conditioned latent transport supervision

The core of the method is a time-conditioned velocity field $v_\theta(z_t, t, s)$ that transports latents along a stochastic bridge. At a randomly sampled time $t \sim \mathcal{U}(0,1)$, the bridge state is

$$z_t = (1-t)z_0 + t\tilde{z}_1 + \sigma\sqrt{t(1-t)}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{3}$$

and the target velocity is the derivative of the transport:

$$u_t = \tilde{z}_1 - z_0 + \sigma\frac{1-2t}{2\sqrt{t(1-t)}}\,\epsilon. \tag{4}$$

The flow-matching objective supervises the network to predict this velocity:

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{z_0,\tilde{z}_1,t,\epsilon}\,\rho\big(v_\theta(z_t,t,s), u_t\big). \tag{5}$$

Here ρ is the pointwise velocity penalty. The implementation supports MSE, Huber, and L_1 variants in the direct velocity-regression path. Equation (5) should therefore be read as the parent bridge formulation. The current Distinct5 tokenizer family emphasized later in the paper resolves instead to the OMF objective path, where the active transport term is closer to endpoint-delta prediction than to the standalone random-time residual above. This is exactly why the paper keeps its metric claim narrow: the current evidence favors the OT + W_1-style mainline over isolated endpoint-only pointwise supervision, while the best local transport residual remains an open experiment rather than a settled theorem.

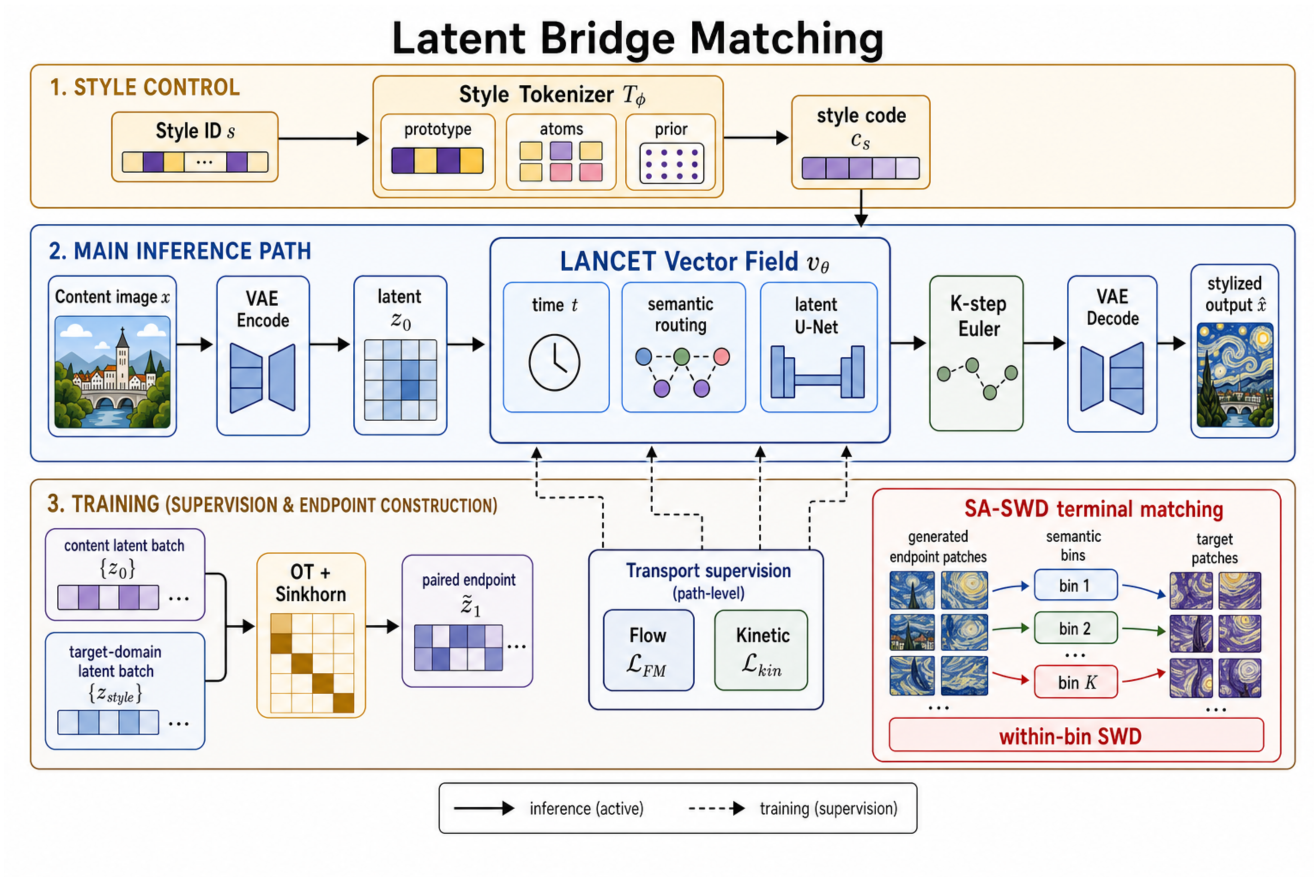

Figure 2: Overview of Latent Bridge Matching. The top band shows style-side control, where the style tokenizer maps the target style identifier into a compact style code. The middle band is the active inference path: content is encoded into a latent, passed through the LANCET vector field with time conditioning and semantic routing, integrated by K-step Euler updates, and decoded into the stylized output. The bottom band shows training-only constraints: target-domain latents enter through mini-batch OT endpoint construction and bridge-style supervision, while the SA-SWD inset makes explicit how semantic bins define terminal within-bin SWD matching. Target-domain latents are therefore training-side supervision only, not inference-time conditioning inputs.

Architecture. The time-conditioned backbone uses sinusoidal time embeddings added to the style code. The implementation is a U-Net-like latent trunk with four semantic cross-attention body blocks at 16×16 resolution. The target style is represented by a learnable spatial prior $P_s \in \mathbb{R}^{128\times16\times16}$. Semantic cross-attention injects the style prior into content features, while skip connections preserve spatial structure.

Style tokenizer as executable control

The style-side interface is deliberately separated from the latent renderer. In the main domain-level protocol, the tokenizer receives only a target style identifier and learned style parameters available at inference:

$$c_s = T_\phi(s), \qquad v_\theta = F_\theta(z_t, t, c_s). \tag{6}$$

It does not read a target image or target latent from the current evaluation batch. Target-domain latents remain on the loss side of the graph through OT coupling and SA-SWD, but they are not conditioning evidence for T_ϕ. This boundary is important: otherwise training would solve a reference-guided problem while evaluation reports a style-id problem. In this sense, T_ϕ names where transport should go, while F_θ decides how much latent motion is actually needed to get there. Throughout this paper, “style tokenizer” means a style-id-conditioned control map, not an image-token or diffusion tokenizer and not a target-image encoder.

The simplest tokenizer is a per-style code table. Our current implementation generalizes this interface without changing the LANCET consumer:

$$c_s = p_s + \gamma \sum_{k=1}^{K} \alpha_{s,k} a_k, \qquad \alpha_s = \mathrm{softmax}(\ell_s/\tau), \tag{7}$$

where p_s is an optional direct prototype, $\{a_k\}$ are shared style atoms, and γ controls the residual atom contribution. Other tested variants replace p_s with class-local prototypes or use a purely shared VQ-style codebook. The tokenizer therefore exposes a measurable style vocabulary while preserving the single-code interface needed by the existing renderer.

For execution, the backbone may also use style spatial priors and content-conditioned routing. These modules modulate where a target-style control is applied, not what target style is requested. This distinction matches the empirical diagnosis: within the tested Distinct5 tokenizer variants, increasing tokenizer capacity alone did not break the style ceiling, while prototype-aware target queues and content-guided routing improved the executed style-content frontier. We therefore phrase tokenizer evidence at the level of *executable representation* in this setup: code geometry alone is insufficient, and style-side distinctions matter only insofar as generated latent residuals remain distinguishable after the frozen or jointly trained LANCET consumes the code.

Complementary roles. The components address distinct subproblems. OT coupling provides target-style endpoints for unpaired supervision, but not transport dynamics. Transport supervision learns how to move toward those endpoints, but by itself does not guarantee that the integrated endpoint matches target-domain texture statistics. SA-SWD regularizes this endpoint mismatch, while semantic cross-attention provides local routing for where style is written and which target patch axes drive the terminal distribution match. The destructive ablations directly verify the roles of terminal distribution matching and kinetic regularization; the landed matched Random-SWD versus SA-SWD control does not show a clean positive superiority result for semantic projection-axis selection beyond ordinary endpoint SWD on the current Distinct5 packet.

Semantic-aligned terminal distribution matching

In addition to flow-matching supervision, a terminal loss regularizes the Euler-integrated endpoint toward the target style distribution. Let $\Phi_\theta(z_0, s)$ denote the endpoint obtained by integrating v_θ from z_0 with style s. The terminal loss is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1\big(\Phi_\theta(z_0, s),\ \tilde{z}_1\big), \tag{8}$$

where the SWD compares generated endpoint patches with coupled target patches. In the current mainline configuration, we use semantic-aligned rather than purely random projections. Let K_s denote the keys produced by semantic cross-attention for target style s. We rank spatial target positions by the average magnitude of K_s and use the corresponding target latent patch vectors as normalized projection directions. This defines the SA-SWD design used in the current mainline: terminal distribution matching is still computed by sorted one-dimensional projections, but the projection axes are selected from the same semantic routing signal that injects style into the backbone.

The complete training objective combines transport supervision, terminal distribution matching, and kinetic path regularization:

$$\mathcal{L} = \mathcal{L}_{\text{transport}} + \lambda_{\text{term}}\mathcal{L}_{\text{term}} + \lambda_{\text{kin}}\mathcal{L}_{\text{kin}}, \tag{9}$$

where $\mathcal{L}_{\text{transport}}$ is instantiated either as the bridge-formulation penalty in Eq. (5) or as the current OMF-style endpoint-delta transport term, depending on the resolved objective family. where $\mathcal{L}_{\text{kin}} = \mathbb{E}\|v_\theta\|_2^2$ penalizes excessive latent displacement. The primary configuration uses $\lambda_{\text{term}} = 20.0$ and $\lambda_{\text{kin}} = 1.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This three-part loss separates the forces that drive style from those that preserve content.

Metric choice in latent space. The training objective does not treat all latent discrepancies as Euclidean reconstruction errors. The kinetic term uses an L_2 penalty on velocity because it represents path energy, while terminal endpoint matching uses Wasserstein-1 style discrepancies over sorted projections. This separation is important in compressed VAE latents: pointwise endpoint MSE would reward mean solutions and over-penalize semantically valid local rearrangements, whereas SA-SWD preserves distributional supervision without requiring paired spatial correspondence. The current reproduced Distinct5 checkpoints do *not* enforce style alignment through latent endpoint MSE. Their style pressure is instead carried by OT endpoint construction plus W_1-style terminal matching, which appears more consequential than endpoint-only pointwise supervision in the current measured setting. We therefore treat the unresolved choice of local transport penalty as an experiment-design question, not as a closed headline claim.

Theoretical grounding of latent transport. The transport design admits a lightweight design-grounding analysis (Sec.): Theorem 1 connects the training kinetic loss to the inference-time path action, Theorem 2 bounds the Euler discretization error at $O(1/K)$, and Theorem 3 provides a Lipschitz-continuity rationale for OT-based endpoint coupling. Each result is paired with empirical checks where available.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^M$, the empirical terminal objective is

$$\begin{aligned}\mathcal{L}_{\text{swd}} = \sum_{r\in\mathcal{R}} \frac{1}{M}\sum_{m=1}^{M} \Big\| &\text{sort}\big(\mathcal{P}_r(\hat{z}_1)\theta_m\big) \\ &- \text{sort}\big(\mathcal{P}_r(z_s)\theta_m\big)\Big\|_1.\end{aligned} \tag{10}$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics along semantic routing directions, not whether it matches any particular artwork spatially.

Formal analysis

We provide three formal results that characterize key properties of the proposed method. Full proofs appear in the supplement. For brevity we state the results and their empirical support.

Notation. Recall the OMF training regime: $\hat{v} = v_\theta(z_0, 1, s)$, $\hat{z}_1 = z_0 + \hat{v}$. At inference, integration follows $dz/dt = v_\theta(z, t, s)$ with K-step Euler: $z_{k+1} = z_k +$

$\Delta t\, v_\theta(z_k, t_k, s)$ where $\Delta t = 1/K$, $t_k = (k+0.5)\Delta t$. Define the path action $\mathcal{A}(v) = \int_0^1 \mathbb{E}\|v_\theta(z(t), t, s)\|^2\, dt$ and the training kinetic loss $\mathcal{L}_{\text{kin}} = \mathbb{E}\|v_\theta(z_0, 1, s)\|^2$.

Scope and limitation. Theorem 1 should be read as a local design justification, not as a global characterization of arbitrary learned trajectories. The result assumes bounded temporal drift and Lipschitz continuity, and therefore explains when endpoint kinetic regularization is a useful proxy for path energy. It does not claim that a single endpoint penalty exactly controls the full trajectory in all regimes. This is why we pair the theorem with empirical path-statistics measurements rather than presenting it as a closed-form solution of the global latent transport problem.

Theorem 1 (Path action decomposition). Assume that for a fixed content z_0 the velocity norm varies at most δ with t, i.e., $\sup_t \big|\|v_\theta(z_0, t, s)\| - \|v_\theta(z_0, 1, s)\|\big| \leq \delta$, and that v_θ is L-Lipschitz in z. Let M bound $\|v_\theta\|$ along the trajectory. Then the path action decomposes as

$$\mathcal{A}(v) = \mathcal{L}_{\text{kin}} + \varepsilon_t + \varepsilon_z, \tag{11}$$

where $|\varepsilon_t| \leq 2\delta\sqrt{\mathcal{L}_{\text{kin}}} + \delta^2$ and $|\varepsilon_z| \leq LM$. If δ and LM are small, the inference-time path energy is approximated by the training kinetic loss.

Empirical support. Experimentally (128-step integration, D0 model): $\mathcal{L}_{\text{kin}} \approx 1017$, $\mathcal{A}(v) \approx 698$, yielding $|\varepsilon_t + \varepsilon_z| \approx 319$. Removing $\mathcal{L}_{\text{kin}}$ (D2) reduces the path length ratio from 0.98 to 0.94, confirming the kinetic term controls path regularity.

Theorem 2 (Euler discretization bound). Under the L-Lipschitz condition on v_θ, the K-step Euler integration error satisfies

$$\|z(1) - z^{(K)}\| \leq \frac{C}{K}, \qquad C = \frac{1}{2}(ML_t + M^2 L)\,\frac{e^L - 1}{L}, \tag{12}$$

where L_t bounds the t-derivative of v_θ.

Empirical support. Measured over 160 content latents (D0 model, 256-step reference): $\|z^{(K)} - z^{(256)}\| = \{10.7, 3.80, 1.74, 0.84, 0.40, 0.19, 0.084, 0.027\}$ for $K = 1, 2, 4, 8, 16, 32, 64, 128$; the error halves when K doubles, confirming the $O(1/K)$ scaling. At $K = 12$ (the default), the error is < 0.6—negligible relative to the style-transfer effect.

Theorem 3 (OT assignment continuity). Let the SWD cost $c(\cdot,\cdot)$ be L_c-Lipschitz and let $\pi^\star$ be the ε-Sinkhorn plan on a batch of size B. Let D denote the diameter of the target latent set. Then, for any two content latents z_0, z_0', the expected matched targets satisfy

$$\big\|\mathbb{E}_{\pi^\star}[\,\tilde{z}_1 \mid z_0\,] - \mathbb{E}_{\pi^\star}[\,\tilde{z}_1 \mid z_0'\,]\big\| \leq \frac{D \cdot L_c}{\varepsilon}\,\|z_0 - z_0'\| + O(\varepsilon). \tag{13}$$

Thus the OT assignment map is (DL_c/ε)-Lipschitz: nearby content latents receive nearby matched targets. Random uniform coupling provides no such guarantee. The Lipschitz property implies smoother supervision gradients and more directionally coherent displacement vectors across the batch.

Empirical support. Compared over 640 content latents: OT coupling yields mean cosine similarity to the batch-mean direction of 0.150, vs. 0.124 for random coupling, a 21% improvement. The transport cost is also reduced by 4.2% (0.119 vs. 0.124).

Experiments

Protocol hierarchy and metrics

We organize the evidence into two non-interchangeable result families, with Distinct5-512 as the primary stress benchmark and the historical strict-750 packet as contextual support. This separation is necessary because the same scalar pair, CLIP-style and LPIPS, can be misleading when datasets, image resolution, or baseline training protocols change.

Historical strict-750. The main legacy protocol uses five domains—photo, Hayao, Monet, Van Gogh, and Cezanne—and evaluates all 5×5 source-target directions with 30 source images per source domain, for 750 generated outputs. This is the protocol used for comparisons to SaMST, S2WAT, StyleID, AdaIN, StyTr2, CAST, AesFA, and AesPA-Net.

Distinct5-512 stress split. The new stress split uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These five classes were chosen as the most strongly separated subset under a CLIP-style screening pass over WikiArt categories, rather than being hand-picked for LBM. Evaluation again uses all 5×5 directions over 750 generated images. This split complements the historical table with a stronger domain-separation test and an explicit identical-image baseline.

We report CLIP-style (CLIP-S) and CLIP-content (CLIP-C) using CLIP ViT-B/32 embeddings (Radford et al. 2021), together with LPIPS-content and

$$\text{EC} = \text{CLIP-style} \cdot (1 - \text{LPIPS}). \tag{14}$$

For Distinct5-512 we additionally report style gain over the unchanged-image reference,

$$\Delta_{\text{idt}} = \text{CLIP-S(method)} - \text{CLIP-S(idt)}, \tag{15}$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. We use this quantity to define the metric-stress regime exposed by Distinct5-512: raw CLIP-style should not be treated as sufficient evidence of successful target-style transfer if a method either fails to exceed the identical-image prior ($\Delta_{\text{idt}} \leq 0$) or exceeds it only by moving into a clearly damaged LPIPS region while also deteriorating broad artifact diagnostics. In this paper, CLIP-S is interpreted as target-style affinity, LPIPS as content displacement, and ArtFID as a broader art-domain realism and artifact diagnostic rather than a direct style-gain score. In particular, lower ArtFID alone does not certify useful stylization on separated art-to-art transfer: its content term rewards source preservation, and its distributional term can reward remaining in a plausible target-domain art manifold even when the requested target-style movement is still below the unchanged-image baseline. EC is used only as a Pareto summary; claims are checked against CLIP-S and LPIPS individually. For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, shallow Gram micro statistics, and KID where available. All strict-750 baselines use the same source images, target domains, preprocessing, and identity directions. For arbitrary-style methods, one target-domain reference image is supplied per target style; for style-id LBM, the model receives only the target domain identifier. The 'idt' control is introduced as a diagnostic baseline for art-to-art transfer rather than as a community standard: our claim is that Distinct5 makes explicit the role of the unchanged-image prior on this split, not that any single existing metric should be discarded.

Distinct5-512 stress benchmark

Distinct5-512 is a metric-stress benchmark rather than a universal art benchmark. It is not a contrived out-of-distribution trap or a failure-mined adversarial set: all images come from standard WikiArt classes, and the five domains are selected only because they are among the most separated under a CLIP-style screening pass. Table 1 gives the exact numbers. The comparison is stricter than a single target direction: every method must handle identity directions, nearby art-to-art transfers, and distant cross-style transfers. That is not an exotic demand; it is the minimum sanity condition for an art-to-art metric. If a method is asked to move between deliberately separated WikiArt styles, unchanged output should not be allowed to masquerade as success. On this split, however, the *idt* baseline already reaches CLIP-S 0.6801 while doing no transfer at all, so raw CLIP-style alone overstates performance. Under the current Distinct5-512 full 5×5 / 750-output protocol, LBM occupies the strongest measured CLIP-S / LPIPS trade-off among the currently reproduced positive-Δ_{idt} operating points. The content-preserving LBM-F checkpoint reaches CLIP-S 0.6969 and LPIPS 0.3186. The timing entries in Table 1 are recorded operating-point wall clocks and are included as context only, not as normalized time-to-parity evidence. SaMST-512 shows a different failure mode: raw style rises to 0.7247, but LPIPS degrades to 0.6255 and targetwise ArtFID worsens to 395.7. The closed e5/e15 packet is already near a plateau on transfer CLIP-S and LPIPS: the two checkpoints differ by less than 0.004/0.002, while e15 lowers targetwise ArtFID from 465.7 to 444.5 and still remains in a high-damage regime. The style-seeking LBM-K checkpoint raises CLIP-S to 0.7010 while remaining far below that high-damage region. Relative to the unchanged-image reference, LBM remains positive under both scopes: LBM-F gives $\Delta_{\mathrm{idt}} = +0.0168$ full and $+0.0244$ transfer-only, while LBM-K gives $+0.0209$ full and $+0.0312$ transfer-only. At the trusted SaMAM 2250 checkpoint, transfer CLIP-S remains below the idt floor even though targetwise ArtFID improves from 216.5 to 146.1 on the full scope and from 323.7 to 148.2 on the transfer-only scope. This is the sharp failure case: the reproduced model is not merely producing arbitrary failures; it pays distortion budget, reaches a lower-targetwise-ArtFID regime, and still fails to achieve positive target-style gain beyond idt. On Distinct5, lower targetwise ArtFID can therefore reflect source-structure preservation and generic target-domain plausibility while still failing the requested target-style transfer. The critique is not that a CVPR baseline is globally invalid; it is that the common absolute-score protocol can certify the wrong success condition for art-to-art transfer on an ordinary separated WikiArt split. This section should therefore be read as a frontier comparison on the current reproduced points, not as proof that semantic projection-axis selection is already isolated as the decisive cause.

Table 1: Distinct5-512 strict-750 stress benchmark. All rows use the same 5×5 all-pairs evaluation over 750 generated images. 'idt' is the unchanged-image reference copied into every target slot. Δ_{idt} is the full-scope style gain over idt and $\Delta_{\mathrm{idt,tr}}$ is the transfer-only off-diagonal gain. 'tw-ArtFID' denotes targetwise ArtFID. SaMAM is reported only at the trusted 2250-step checkpoint; later checkpoints are excluded from the active manuscript path. SaMAM and SaMST training time is cumulative wall time with evaluation excluded; LBM times are per-checkpoint remote RTX 3060 training wall times.

Method	CLIP-S↑	LPIPS↓	EC↑	tw-ArtFID↓	Δ_{idt} ↑	$\Delta_{\mathrm{idt,tr}}$ ↑	Train
idt (no-op)	0.6801	0.0000	0.6801	216.5	0.0000	0.0000	0
SaMAM 2250	0.5811	0.3538	0.3755	146.1	-0.0990	-0.0877	7.6h
SaMST-512 e15	**0.7247**	0.6255	0.2714	395.7	**+0.0446**	**+0.0558**	5.8h
LBM-F e1	0.6969	**0.3186**	**0.4748**	**122.6**	+0.0168	+0.0244	1.2m
LBM-K e1	0.7010	0.3623	0.4470	157.2	+0.0209	+0.0312	1.2m

Historical strict-750 comparison

We then place the historical strict-750 packet as contextual support for the older comparison surface. Table 2 establishes LBM as the strongest compact content-preserving operating point among the reproduced baselines in this historical strict-750 comparison. StyleID reaches the highest raw CLIP-style but collapses content. SaMST is marginally higher on CLIP-S and CLIP-C, yet LBM improves LPIPS and EC while also winning the artifact-sensitive metrics reported below. Relative to S2WAT and AdaIN, LBM preserves substantially more content at comparable or stronger style. The main result here is therefore a quality-frontier claim: within this reproduced table, LBM delivers the cleanest balance of style strength, content fidelity, and artifact control among the compact retrainable systems in the comparison.

Cost matters in two different ways here. First, Table 3 preserves operating-point wall-clock observations from the reproduced records: 310 s for the reported LBM checkpoint, 6769 s for the reproduced SaMST operating point, and 10600 s for S2WAT. These timings are useful for reporting practical footprint, but they are not normalized time-to-parity evidence because stopping rules differ across methods. Second, inference cost remains practical rather than extreme. LBM is not the fastest feed-forward path in absolute milliseconds per image, since lightweight statistic-alignment models such as AdaIN are faster, but it is still far below StyleID's diffusion-style inference cost. The practi-

Table 2: Historical strict-750 comparison in grouped family format. Higher is better for CLIP-S, CLIP-C, and EC; lower is better for LPIPS. Artifact-sensitive metrics are reported separately in Table 4.

Metric	CNN-based			Transformer-based			Training-free		Latent transport
	AesPA	AesFA	AdaIN	StyTr2	S2WAT	SaMST	CAST	StyleID	LBM
CLIP-S ↑	0.5976	0.6496	0.7130	0.6374	0.7139	0.7194	0.6652	**0.7597**	0.7161
CLIP-C ↑	0.5009	0.5497	0.6990	0.6001	0.7465	**0.8193**	0.5562	0.5519	0.8086
LPIPS ↓	0.8215	0.7392	0.6298	0.6345	0.5263	0.4664	0.7263	0.7497	**0.4514**
EC ↑	0.1067	0.1694	0.2639	0.2330	0.3382	0.3839	0.1821	0.1902	**0.3928**

Table 3: Recorded operating-point training and inference cost on the historical strict-750 protocol. All timings are wall-clock on a single RTX 4070 Laptop GPU. Training time is reported to the selected operating point used in this paper and should be read as an operating-point record rather than normalized time-to-parity; inference reports 750-image end-to-end runtime and milliseconds per image.

Method	Train (s)	Infer-750 (s)	ms/img	Params
LBM	**310**	85.4	114	**3.91M**
SaMST	6769	39.8	53	∼6M
S2WAT	10600	–	–	∼7M
CAST	1760	75.5	101	7.01M
StyleID	0	603.3	804	–
AdaIN v32k	9220	**9.3**	**12**	–

Table 4: Artifact-sensitive comparison among high-quality methods.

Metric	Ours e7	SaMST	S2WAT
MUSIQ ↑	49.2059	36.0950	36.5256
MANIQA ↑	0.4057	0.3139	0.1754
DISTS-content ↓	0.2477	0.2943	0.2942
HF-Patch-KID ↓	4.1694	6.7598	12.6623
FFT slope error ↓	0.5473	1.0536	0.7017
Gram micro ↓	0.0798	0.0947	—

Gram micro was not recomputed for S2WAT in the current evaluation protocol.

cal conclusion is therefore procedural rather than universal: the reproduced operating-point records indicate a manageable practical footprint under the present protocol, without constituting normalized time-to-parity evidence.

Artifact-sensitive diagnosis

The qualitative failure modes are clearest on the historical 256-resolution packet, where matched source-target grids and centered texture crops make the surface trade-off visible. SaMST is the closest efficient baseline under the core metrics, but these comparisons show muddy, grain-like surface artifacts that can coexist with competitive CLIP-content and LPIPS because low-frequency object boundaries remain visible. LBM is less aggressive in raw style, but visually cleaner on the same packet.

Table 4 shows that LBM is not merely winning by leaving the image unchanged. Relative to SaMST, it improves MUSIQ from 36.10 to 49.21, MANIQA from 0.314 to 0.406, DISTS-content from 0.294 to 0.248, HF-Patch-KID from 6.76 to 4.17, and FFT slope error from 1.054 to 0.547. A paired bootstrap over the 750 per-image scores for Ours gives 95% intervals of $[0.4467, 0.4563]$ for LPIPS and $[0.7091, 0.7234]$ for CLIP-S. The photo-to-art subset ($n = 120$) yields CLIP-S 0.6716 and LPIPS 0.4882; the non-identity subset ($n = 600$) yields CLIP-S 0.6911 and LPIPS 0.4609.

Tokenizer and representation ablations

These ablations make the tokenizer conclusion sharper than a parameter-count story. Class-local prototypes and global atoms did not beat the baseline frontier, so merely enlarging the tokenizer is insufficient. Content-guided spatial routing helped, which indicates that execution location matters. The largest improvement came from a prototype-aware latent target queue, which reduces target-distribution noise before the terminal loss is applied. Finally, content-adaptive atom routing produced the best style score but regressed LPIPS, suggesting that stronger style-side separation can still demand larger executed motion. A plausible next tokenizer direction is therefore to test a two-factor interface rather than a larger undifferentiated code: a target-style carrier that remains distinguishable after execution, plus a content-risk gate that bounds how much of that carrier should be applied.

Non-training probes support the same diagnosis boundary. On the historical e8 checkpoint, tokenizer codes have near-full rank for five styles (effective rank 3.986, mean off-diagonal cosine 0.015), while generated latent residuals are much more aligned (effective rank 3.324, mean off-diagonal cosine 0.725). Residual variance is also content dominated: source-image variance explains 0.9501 of raw generated-delta variance, while target style explains only 0.0283 before source-image removal but 0.5664 after removing per-source-image residual means. A newer matched Distinct5 successor probe reaches the same boundary from the execution side: tokenizer-code geometry maps only moderately into executed geometry, while realized style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. Taken together, these probes show that distinguishable tokenizer codes do not automatically remain distinguishable after content-conditioned execution. They justify evaluating tokenizer quality through executed control rather than code separability alone, but they do not restore H-family continuity or prove that tokenizer-

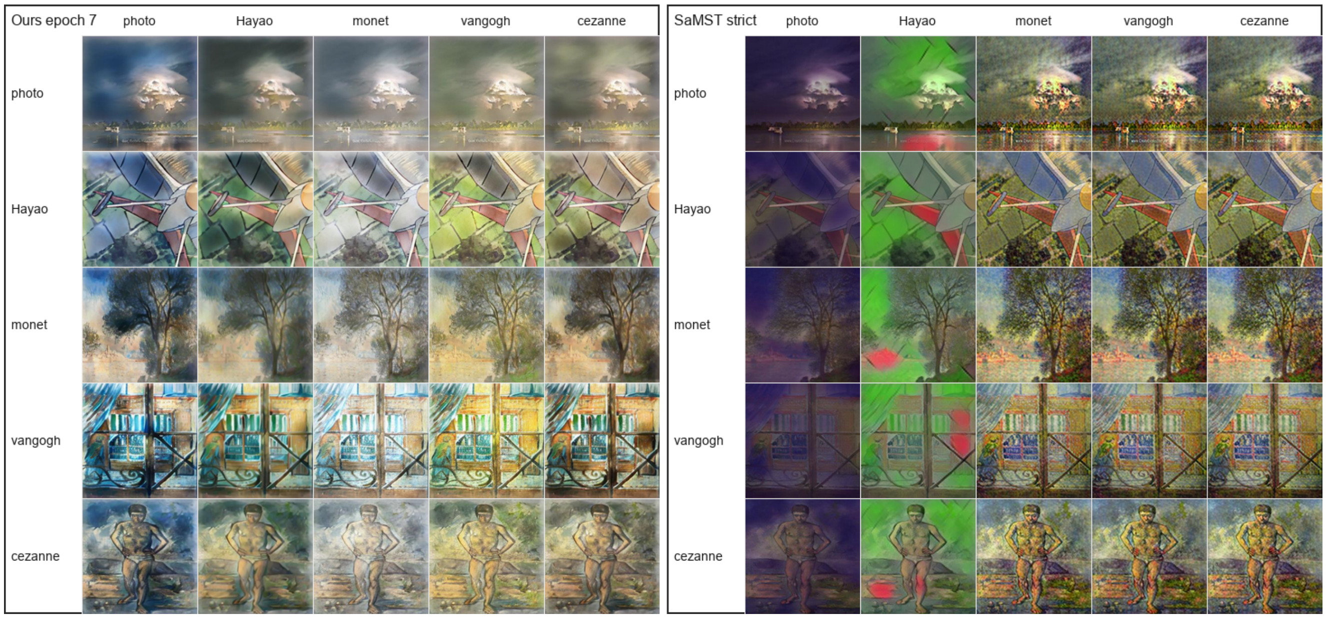

Figure 3: Historical strict-750 qualitative source-target grids for LBM epoch 7 and SaMST strict at 256 resolution. Rows indicate source domains and columns indicate target domains. SaMST preserves coarse structure but often produces muddy, grain-like surface texture, while LBM is visually cleaner.

Table 5: Distinct5-512 tokenizer and representation ablations. The results show that capacity alone is not the issue; target selection, content-guided execution, and sparse routing determine whether the representation is executable.

Variant	Best epoch	CLIP-S	LPIPS	Interpretation
Direct atom residual baseline	8/1	0.6876	0.4468	weak baseline; simple residual atoms do not solve representation
Class-local prototypes	8/1	0.6849	0.4464	more per-class capacity, no better style geometry
Global VQ atoms	8	0.6873	0.4446	small LPIPS gain, no style breakthrough
Content-guided spatial routing	2	0.6907	0.4226	first simultaneous style and LPIPS gain
VQ + content-guided routing	1	0.6898	0.4156	stronger content preservation, style still limited
Prototype-aware latent target queue	1/3	0.6973	0.3331	major jump from better internal target distribution
Annealed prototype queue (F)	1	0.6969	**0.3186**	current LPIPS reference point
Hard-explore prototype queue (H)	2/1	0.6994	0.3213	current balanced point
Content-adaptive VQ atom routing (K)	1	**0.7010**	0.3623	style boost, but larger endpoint movement

Table 6: Destructive ablations under the historical strict-750 protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o SA-SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

side weakness is absent.

Mechanism ablations

Removing terminal distribution matching gives excellent content preservation but weak style, showing that terminal distribution alignment is a primary style driver in the current mainline. Removing kinetic regularization preserves style but collapses content, confirming that path-energy regularization is essential. Strong color loss damages the trade-off, so naive color matching is not used in the mainline. Trajectory straightness analysis gives the same reading: the full model has PLR 0.98, while removing SA-SWD produces near-linear transport to the OT target (PLR 0.9995) and removing kinetic regularization reduces straightness to PLR 0.94 with content collapse.

Sensitivity and efficiency

The 40-run category-weight sweep trained 20 sampling recipes with $K \in \{1, 2\}$ for eight epochs each, evaluating every epoch for 320 total checkpoints. The best EC result was K2 balanced default at epoch 3 with CLIP-S 0.6980, CLIP-C 0.8727, LPIPS 0.3777, and EC 0.4343. The best raw-style result in the sweep was K1 balanced default at epoch 8 with CLIP-S 0.7161, CLIP-C 0.7984, LPIPS 0.4605, and EC 0.3863. The practical reading is simple:

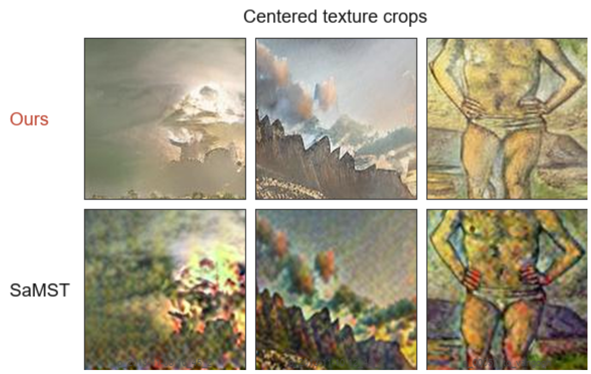

Figure 4: Centered texture crops from matched 256-resolution LBM and SaMST outputs, isolating the local surface-texture failure mode.

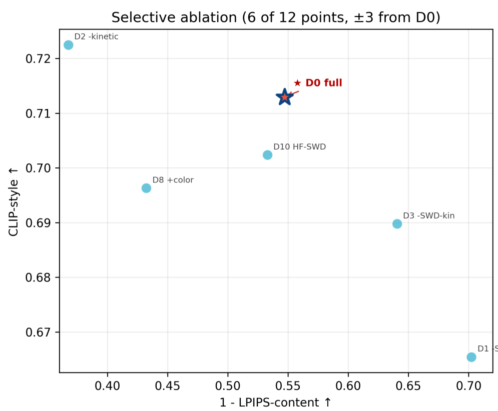

Figure 5: Destructive ablations. Removing SA-SWD improves content but weakens style, while removing kinetic regularization increases style at the cost of severe content degradation.

more complex category weighting was not consistently better than balanced sampling, so we keep this sweep as a supporting sensitivity result rather than spending main-figure budget on it.

Step-count sensitivity is also stable. Evaluating 1, 4, 8, 12, and 16 integration steps on the selected epoch-7 checkpoint changes all-pairs CLIP-S only from 0.7159 to 0.7162, while LPIPS remains near 0.4514. Four steps already recover nearly identical quality, with measured strict-750 throughput increasing from 8.43 images/s at one step to 9.54 images/s at four steps and remaining around 9.34–9.50 images/s for 8–16 steps. A 3×3 grid over $\lambda_{\mathrm{kin}} \in \{0.5, 1.0, 2.0\}$ and $\lambda_{\mathrm{swd}} \in \{10, 20, 30\}$ produced a smooth style-content surface: higher kinetic weight and lighter terminal pressure improve content, while weaker kinetic weight or stronger terminal pressure increases style at the cost of LPIPS.

All reported evaluations use fixed seeds, stored resolved configurations, source snapshots, checkpoints, and per-epoch logs. Historical timings were measured on an RTX 4070 Laptop GPU unless otherwise noted. For Distinct5-512, Figure 6 replaces mixed timing anecdotes with one explicit same-scope packet: cumulative training wall time is plotted with evaluation excluded, and each method family is labeled by timing mode rather than being forced into a false normalized parity claim. For the historical table, training and inference cost are reported explicitly in Table 3 rather than being folded into a single efficiency claim.

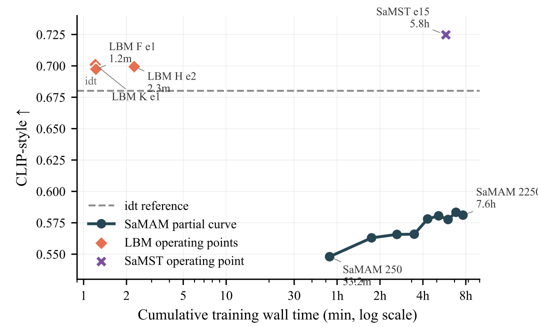

Figure 6: Same-scope Distinct5 timing-context artifact under the reproduced full 5×5 / 750-output protocol, with cumulative training wall time and evaluation excluded from the horizontal axis. LBM is shown as reviewed operating-point records, SaMAM as bounded pre-2250 timing records with the active manuscript boundary fixed at the trusted 2250 checkpoint, SaMST as the currently available operating point, and *idt* as the unchanged-image reference. This figure is used as bounded timing context rather than as a closed normalized time-to-parity claim.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. On the historical strict-750 protocol, StyleID has the highest raw style but collapses content, SaMST is slightly stronger than LBM on CLIP-S and CLIP-C but shows grain-like local artifacts, and LBM occupies a cleaner LPIPS/artifact-sensitive operating point. Distinct5-512 sharpens the same lesson further because its no-op baseline is measured explicitly: higher raw style is achievable, but only by moving into a severe LPIPS and ArtFID failure region, and some baselines fail even to exceed the identical-image prior. This is best read as a bounded evaluation diagnosis, not as a blanket rejection of prior baselines. Earlier AST evaluation work already notes that protocols are inconsistent across papers and that automatic scores can be style-dependent or misaligned with human judgment (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). Our narrower claim is that art-to-art transfer benefits from reporting an unchanged-image reference that measures how much target-style similarity is already present before stylization, especially on separated art-to-art splits. Across both protocols, the consistent conclusion is a scope-pinned frontier result: on Distinct5-512 and within the currently reproduced positive-Δ_{idt} operating points, LBM gives the strongest measured CLIP-S / LPIPS trade-off, while the remaining open mechanism question is how faithfully style-side distinctions survive execution rather than whether code-space changes can be induced at all.

The tokenizer experiments narrow the design question. Within the tested Distinct5 tokenizer variants, a larger style code or more class-local prototypes is not sufficient. The sharper open issue in this family is whether target-style information survives execution through a content-conditioned vector field strongly enough to yield real style gain beyond the unchanged-image reference. The current evidence therefore favors target-style carriers, prototype-aware target queues, and content-aware routing over black-box token capacity in this setting. A landed matched Distinct5 successor probe sharpens that boundary further: tokenizer code separation maps only moderately into executed separation, while style gain over the unchanged-image reference tracks executed movement more closely than raw code geometry. This keeps executed style survival as the sharper mechanism question in the current successor family without restoring H-family continuity or ruling out tokenizer-side weakness. It also shows a trade-off: content-adaptive atom routing can increase style strength but may increase endpoint movement and LPIPS. A plausible next-stage tokenizer direction is therefore to factor target representation from execution budget rather than simply adding parameters.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher resolutions may need better solvers or stronger constraints. Third, SA-SWD is an efficient empirical distribution loss, not an exact global transport solver, and the landed semantic-vs-random axis control does not support a clean positive superiority claim for semantic projection-axis selection on the current Distinct5 packet. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. Finally, Distinct5-512 is a five-domain stress split rather than a universal art benchmark, so broader domain coverage remains future work even though the current cross-method comparison is already informative on this split.

Conclusion

We presented Latent Bridge Matching, an OT-coupled latent flow framework for domain-level artistic transfer. The method separates endpoint construction, path-wise flow learning, semantic terminal distribution matching, and style-tokenized execution. Under the reproduced historical strict-750 protocol, LBM reaches a cleaner style-content trade-off than SaMST, S2WAT, StyleID, and AdaIN-family baselines, with training and inference cost reported explicitly as operating-point records rather than as normalized parity evidence. The Distinct5-512 metric-stress benchmark sharpens the central insight: once the identical-image prior is made explicit, LBM stays above the no-op floor and defines the strongest measured CLIP-S / LPIPS trade-off among the currently reproduced positive-Δ_{idt} points on this separated-domain split. The tokenizer study then narrows the next design target. Within the tested Distinct5 tokenizer variants, current probes, including a landed matched Distinct5 successor probe, show that raw code enlargement is insufficient and that executed style survival through a content-conditioned renderer is the sharper mechanism question, without ruling out tokenizer-side weakness outright. One plausible next representation stage is therefore to factor target-style content from execution budget through an explicit target carrier plus content-risk gate.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides three formal results with proofs and partial empirical support: a path-action decomposition connecting the training kinetic loss to inference-time path energy (Theorem 1), an $O(1/K)$ Euler discretization error bound (Theorem 2), and a Lipschitz-continuity result for the OT assignment map explaining its directional-coherence benefit (Theorem 3). Full proofs are in the supplement. These results justify the three-part design without claiming equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **No**.
- All source code required for conducting and analyzing the experiments is included in a code appendix: **No**.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial**.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Partial**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.